

CLASSIFICATION: OFFICIAL / RESTRICTED

Comprehensive Project Proposal & Technical Architecture Report

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Bangladesh Citizen Hub (BCH)

A Unified National Ecosystem for Citizen Safety, Decentralized Education, & Civic Engagement Framework

বাংলাদেশ সিটিজেন হাব

"One Identity. One Platform. One Safe Nation."

"এক পরিচয়, এক প্ল্যাটফর্ম, একটি নিরাপদ বাংলাদেশ।"

Proposed Framework for Strategic Integration under the Guidance of:
Information and Communication Technology (ICT) Division
& **Ministry of Home Affairs**
Government of the People's Republic of Bangladesh

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1 Executive Summary | নির্বাহী সারসংক্ষেপ

1.1 English Context

The **Bangladesh Citizen Hub (BCH)** is a unified, next-generation national digital ecosystem designed to integrate citizen safety, decentralized education, structural skill development, public grievance/civic complaint management, missing person location tracking, and real-time emergency response infrastructure into a singular secure portal.

Rather than deploying isolated digital silos, BCH establishes a high-integrity, multi-tier identity-verified network. This framework bridges structural gaps between citizens, municipal authorities, law enforcement agencies, and educational institutions to foster national development, digital transparency, and public security.

1.2 বাংলা সারসংক্ষেপ

Bangladesh Citizen Hub (BCH) একটি সর্বজনীন ও সমন্বিত ডিজিটাল প্ল্যাটফর্ম, যার মাধ্যমে নাগরিক নিরাপত্তা, শিক্ষা, দক্ষতা উন্নয়ন, নাগরিক সমস্যা রিপোর্টিং, নিখোঁজ ব্যক্তি সহায়তা এবং জরুরি রেসপন্স সিস্টেমকে একটি একক ইকোসিস্টেমে রূপান্তর করা হবে। এই প্ল্যাটফর্মের মূল লক্ষ্য হলো একটি অত্যন্ত বিশ্বস্ত এবং যাচাইকৃত নাগরিক নেটওয়ার্ক (Verified Citizen Network) তৈরি করা, যা বিচ্ছিন্ন অ্যাপ্লিকেশনের সীমাবদ্ধতা দূর করে দ্রুত জরুরি সহায়তা, স্বচ্ছ পাবলিক সার্ভিস অ্যাকাউন্টেবিলিটি এবং সবার জন্য সমান শিক্ষার সুযোগ নিশ্চিত করবে।

2 Vision, Mission & Objectives | ভিশন ও মিশন

- **Vision / ভিশন:** To build a safe, highly skilled, and digitally empowered sustainable society. একটি নিরাপদ, দক্ষ এবং দীর্ঘস্থায়ী প্রযুক্তিনির্ভর ডিজিটাল সমাজ গঠন করা।
- **Mission / মিশন:** To provide every citizen with a trusted digital profile that grants seamless access to learning resources, real-time safety nets, and local civic participation. প্রত্যেক নাগরিকের জন্য একটি নির্ভরযোগ্য ডিজিটাল পরিচয় নিশ্চিত করা এবং শিক্ষা, নিরাপত্তা সরঞ্জাম, নাগরিক অংশগ্রহণ ও সামাজিক সেবার অধিকার উন্মুক্ত করা।

2.1 Core Institutional Objectives

1. **Citizen Safety:** Minimize incident response latency during physical emergencies.
2. **Education Equalization:** Provide high-tier, curriculum-aligned academic resources nationwide.
3. **Workforce Readiness:** Build specialized micro-learning and technical tracks to improve youth employment metrics.
4. **Civic Transparency:** Implement transparent geo-tagged community problem reporting frameworks.

3 Core Architectural Layout | মূল প্ল্যাটফর্ম কাঠামো

The platform's features are compartmentalized into six distinct, interconnected service engines.

4 MODULE 1: Verified Citizen Identity System | যাচাইকৃত নাগরিক পরিচয়

To eliminate malicious actors and optimize civil accountability, the system implements strict registration parameters alongside an electronic Know-Your-Citizen (e-KYC) verification workflow.

4.1 Registration Requirements & Field Schemas

- **Demographics:** Full Name (English & বাংলা), Date of Birth (DoB), Gender, and Nationality.

Table 1: System Module Framework Matrix

Module ID	Module Name	Primary Functional Domain
Module 1	Verified Citizen Identity System	Multi-factor verification, e-KYC, and trust validation.
Module 2	Education Resource Hub	Academic preparation, curriculum libraries, and AI tutoring.
Module 3	Skill Development Hub	Targeted technical tracks, certifications, and career roadmaps.
Module 4	Civic Complaint System	Geo-mapped local issue logging, tracking, and escalations.
Module 5	Missing & Found Network	Crowdsourced search parameters and automated matching.
Module 6	Emergency SOS & Safety Infrastructure	Offline panic alerting, trip monitoring, and agency routing.

- **Contact Telemetry:** Mobile Number (Validated via time-restricted One-Time Password - OTP), and Email Address.
- **Government Identification Vectors:** National Identification Number (NID) or Digital Birth Registration Number, supported by dual-side optical document uploads.
- **Granular Geographic Addr. Fields:** Division, District, Upazila, Union/Ward, and Village/Street coordinates for both Present and Permanent residences.
- **Critical Medical Profiling:** Mandatory Blood Group log, chronic allergies, underlying medical conditions, and explicit emergency treatment instructions.

4.2 Biometric Face Verification System

To protect profiles against identity theft, registration requires a live-capture biometric check:

$$\text{Biometric Match Probability } (P_{\text{match}}) \geq \tau_{\text{threshold}}$$

The system confirms identity via automated blink checks, 3D parallax head movement detection, and live image comparison against uploaded government documents.

4.3 Verification Progression Hierarchy

- **Level 1 [Mobile Verified]:** OTP confirmed. Basic network read access.
- **Level 2 [Email Verified]:** Communication node authenticated.
- **Level 3 [Identity Verified]:** Government database NID/Birth certificate cross-check match successful.
- **Level 4 [Face Verified]:** Biometric verification confirms the user matches their documentation.
- **Level 5 [Fully Verified Citizen]:** Complete system clearance. Profile displays the *Citizen Verification Badge*.

5 MODULE 2: Education Resource Hub | শিক্ষা রিসোর্স প্ল্যাটফর্ম

This module democratizes access to academic assets to provide high-quality learning resources to users throughout the country.

5.1 Functional Resource Architecture

- **Learning Content Repositories:** Open-access libraries containing syllabi, curated text chapters, and structured question banks.

- **Exam Preparation Portals:** Dedicated environments for national and public service exams: Secondary School Certificate (SSC), Higher Secondary School Certificate (HSC), Polytechnic Diplomas, University Admissions, and Bangladesh Civil Service (BCS) / Government Job preparation profiles.
- **Gamified Assessment Engine:** Daily automated quizzes, adaptive mock tracking examinations, and localized/national performance ranking matrices.

5.2 Linear Learning Progression Spec (e.g., Web Development)

HTML → CSS → JavaScript → React Ecosystem → Backend Architectures → Cloud Deployment

5.3 AI Academic Assistant (Planned Integration Phase)

The platform will include an LLM-driven academic assistant capable of real-time multi-turn concept explanations, automated study planning based on performance data, and localized text summarizations.

6 MODULE 3: Skill Development Hub | দক্ষতা উন্নয়ন প্ল্যাটফর্ম

This framework focuses on training targeted demographics to improve national workforce deployment.

6.1 Target Training Domains

- **Information Technology:** Software programming paradigms, network administration, artificial intelligence models, and cybersecurity workflows.
- **Vocational Healthcare Support:** Specialized caregiving tracks, basic paramedic fields, and patient support training.
- **Global Language Curricula:** Structured courses for languages with high international employment demand, including English, Japanese, and Korean.
- **Digital Freelancing Ecosystems:** Practical guides for navigating platforms like Upwork and Fiverr, managing client relationships, and accessing remote jobs.

6.2 Operational Pipeline

The hub maps user skill profiles directly against an integrated institutional directory of certified private and government training facilities. Users can display their course achievements using a tamper-proof digital certification showcase.

7 MODULE 4: Civic Complaint System | নাগরিক সমস্যা রিপোর্টিং

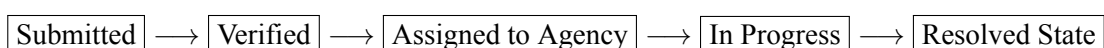
This module provides a direct, transparent channel for reporting localized municipal and infrastructural issues.

7.1 Reporting Vectors & Categories

- **Infrastructure:** Damaged roads, compromised bridge networks, and dangerous footpaths.
- **Utilities Status:** Grid disruptions, water quality alerts, and gas line leaks.
- **Public Safety Risks:** Malfunctioning street lighting or unlit areas with elevated safety risks.
- **Environmental Concerns:** Mismanaged municipal waste piles, illegal pollution sources, and localized waterlogging incidents.

7.2 Workflow & Resolution Tracking Pipeline

Every submission requires a geo-tagged photo or video upload. The system automatically reads the device's GPS telemetry data to plot the issue on an administrative heatmap. Local users can upvote nearby reports to help municipal teams prioritize high-impact issues.



8 MODULE 5: Missing & Found System | নিখোঁজ ও হারানো সহায়তা

This module functions as a centralized, searchable registry for missing persons, lost critical documentation, and high-value personal property.

8.1 Operational Parameter Configurations

Table 2: Missing & Found Categorization Framework

Target Classification	Data Anchors Required	System Response Action
Missing Individuals	Portrait, Height, Age, Last Known GPS	Emergency broadcast to area volunteers
Identity Documents	NID/Passport Serial, Scan Copy	Registry matching for recovery flags
High-Value Property	Device IMEI, Hardware Serial Hash	Automated database match evaluation

The platform uses facial recognition algorithms to analyze uploaded images of found or unidentified individuals against active missing person profiles, allowing teams to quickly notify emergency contacts upon a successful match.

9 MODULE 6: Emergency SOS & Safety System | জরুরি নিরাপত্তা ব্যবস্থা

This module serves as the primary system for real-time safety monitoring and active crisis intervention.

9.1 Guardian Mode & Dead Man's Switch Engine

When traveling or entering higher-risk settings, users can register an active journey log containing their intended destination, the contact details of any meeting partners, and a strict expected return timestamp (T_{return}).

The platform requires periodic check-ins. If a user fails to confirm their safety before the check-in window closes ($t > T_{\text{return}} + \Delta t$), the system automatically executes its emergency protocol.

9.2 The SOS Panic Event Chain

Activating the single-tap Panic SOS trigger initiates the following automated pipeline:

1. Captures high-precision GPS coordinate arrays.
2. Transmits localized text alerts to the user's pre-configured primary emergency contacts.
3. Begins recording ambient audio through the device microphone as encrypted telemetry.
4. Queues the user's profile data for immediate routing to law enforcement dispatch systems.

9.3 Zero-Connectivity Offline Emergency Protocol

If cellular data networks are unavailable, the platform shifts to an offline protocol. It compresses critical safety tokens into standard SMS formats and routes them through GSM channels to the user's emergency contacts, while storing localized coordinates in cache memory until connectivity is restored.

10 Proposed Government & Emergency Integration Architecture

Critical Integration Requirement:

Full deployment of Section 10's operational pipeline is subject to regulatory approval, data exchange clearances, and integration protocols from the Ministry of Home Affairs and the ICT Division.

The system is designed to interface directly with national emergency dispatch centers (such as the National Emergency Service 999, Bangladesh Police units, Rapid Action Battalion, and Fire Service personnel) via a secure API gateway.

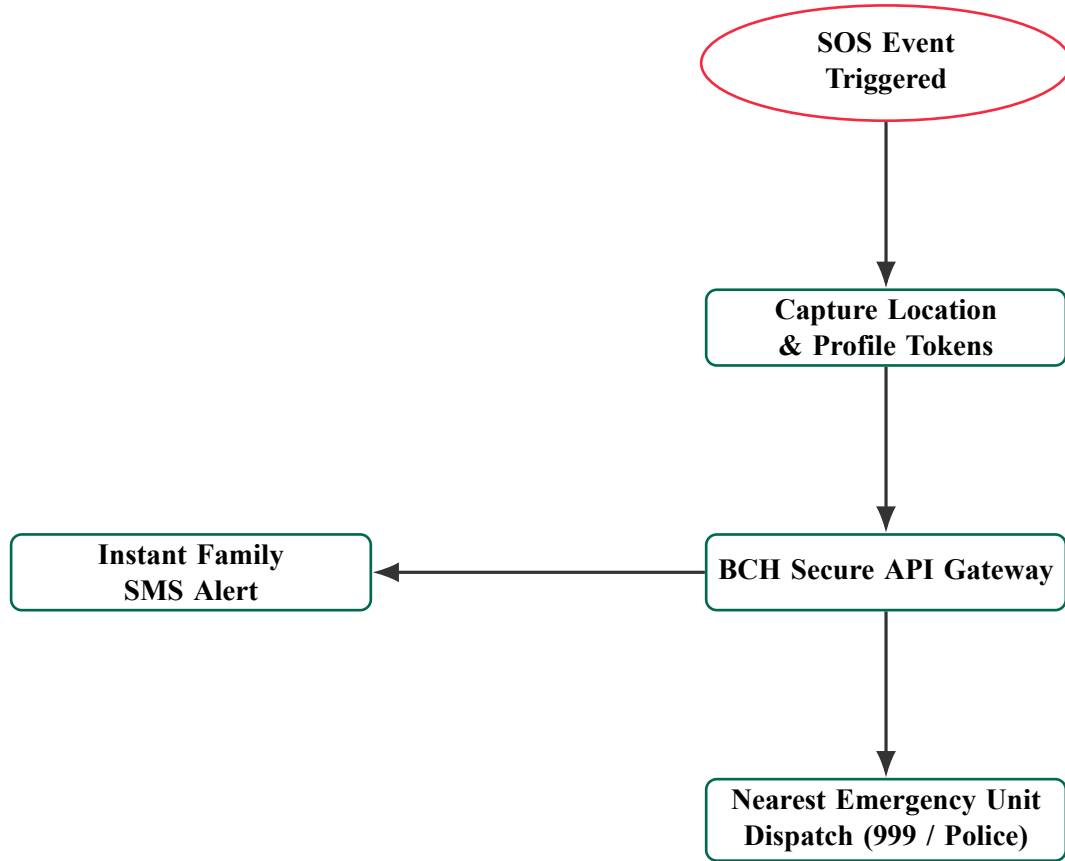


Figure 1: Proposed Emergency Data Orchestration Topology

11 Security, Privacy & Data Governance Framework

To safeguard sensitive user data, the platform implements a comprehensive security architecture:

- **Cryptographic Protocols:** All personal identity files, NID scans, and location data are protected using AES-256 encryption at rest and TLS 1.3 protocols during transmission.
- **Access Control Architecture:** The system uses Role-Based Access Control (RBAC). External municipal entities can only view anonymized, aggregated civic problem data, while full emergency profiles are only unlocked for authorized dispatch personnel during active SOS events.
- **User-Centric Privacy Controls:** Citizens can adjust their profile visibility settings and choose exactly when and how their location information is shared through the platform.

12 Projected National Impact Metrics

- **Public Safety:** Provides emergency dispatch teams with precise location data, helping reduce response times during critical incidents.
- **Educational Access:** Helps bridge the urban-rural opportunity gap by providing standardized study materials to students nationwide.
- **Economic Support:** Connects users with structured training directories and clear career paths to help prepare the youth workforce for digital employment.
- **Administrative Insights:** Offers local government agencies clear, data-driven visibility into regional infrastructure and utility needs.

13 Conclusion | উপসংহার

The **Bangladesh Citizen Hub** serves as a foundational digital infrastructure project that aligns with national digital development strategies. By combining emergency support networks, accessible learning resources, and transparent civic reporting tools into a single identity-verified platform, BCH provides an effective, secure framework for community empowerment and public safety.

Bangladesh Citizen Hub (BCH) কোনো সাধারণ বিচ্ছিন্ন ওয়েব অ্যাপ্লিকেশন নয়; এটি একটি দীর্ঘমেয়াদী ও জাতীয় পর্যায়ে নাগরিক ইকোসিস্টেম। এটি একদিকে যেমন নাগরিকদের জীবন ও সম্পদের নিরাপত্তা নিশ্চিত করবে, অন্যদিকে তেমনি শিক্ষা ও দক্ষতা উন্নয়নের মাধ্যমে একটি দক্ষ মানবসম্পদ তৈরিতে অবদান রাখবে। সরকারি ও প্রাতিষ্ঠানিক নীতিমালার সমন্বয়ে এই প্ল্যাটফর্মটি স্মার্ট বাংলাদেশ বিনির্মাণে এবং একটি নিরাপদ, স্বচ্ছ ও ডিজিটাল সংযুক্ত সমাজ গঠনে গুরুত্বপূর্ণ ভূমিকা পালন করবে।